

Eigenvalues and inverse problems

The Euclidean distance degree of the real symplectic group

Difficulty: challenging **Importance:** interesting to specialist**Status:** Partially resolved

Rating rationale: Challenging reflects an all-ranks algebraic critical-point count with only low-rank computations; specialist impact concerns the algebraic complexity of symplectic matrix nearness.

Supporting Stein reduction and critical-point certificates — 14 September 2026

Sidney Holden (Center for Computational Biology, Flatiron Institute, Simons Foundation). The [report](#), Sections 3–8, supplies the reciprocal-eigenvalue denominator factorization, Stein normal form, Euler-characteristic and parity arguments, and zero-coupling boundary analysis. These supporting results do not evaluate the generic degree in all ranks. The submitted certificates report 543 simple rank-three roots and 31,183 rank-four roots; parity yields lower bounds 544 and 31,184, not completeness. The independent audit distinguishes complete low-rank reruns from sampled rank-four checking. The new all-rank parity theorem is recorded as partial progress. The proposed all-rank formula remains open; status is Partially resolved.

See the [independent informal Codex AI-agent review](#) for the exact audited scope and checks, and the [submission record](#) for verified affiliation, original materials and reproduction details. No full resolution, external human peer review, formal verification or historical novelty is claimed. No Lean verification was performed.

Supporting algebraic reduction — 12 September 2026

Sidney Holden (Center for Computational Biology, Flatiron Institute, Simons Foundation) gives a [skew-multiplier reduction](#) to a saturated quartic system on the regular locus and proves that generic critical points avoid the singular denominator. An [independent informal Codex AI-agent audit](#) passed this reduction. It does not count new generic critical points; the rederived $D_1 = 4$ is already due to Baaijens and Draisma. The all-ranks formula remains open and status stays Open. [Submission and verified affiliation](#). No novelty, external human peer review or formal verification is claimed.

For each $m \geq 1$, set

$$J = \begin{pmatrix} 0 & I_m \\ -I_m & 0 \end{pmatrix}, \quad \mathrm{Sp}_{2m}(\mathbb{C}) = \{X \in \mathbb{C}^{2m \times 2m} : X^T J X = J\}.$$

For a generic data matrix $U \in \mathbb{C}^{2m \times 2m}$, let D_m be the number of complex critical points on this variety of

$$f_U(X) = \mathrm{tr}((X - U)^T(X - U)).$$

This is the complexification of squared real Frobenius distance; the transpose is not a conjugate transpose. Equivalently, count the solutions X of $X^T J X = J$ and

$$\mathrm{tr}((U - X)^T X H) = 0 \quad \text{for every } H \in \mathbb{C}^{2m \times 2m} \text{ satisfying } H^T J + J H = 0.$$

“Generic” means outside a proper Zariski-closed set where the finite critical-point count can change. Is

$$D_m = 2^{m^2} + 2^{2m-1} \quad \text{for every } m \geq 1?$$

Relevance

D_m measures the algebraic complexity of finding the nearest real symplectic matrix. This is a structured matrix nearness problem, relevant when numerical approximations should preserve a symplectic form. The standard Frobenius inner product and the displayed J are part of the problem.

References

- J. A. Baaijens and J. Draisma, *Euclidean distance degrees of real algebraic groups*, Linear Algebra and its Applications 467 (2015), 174–187, §5, p.187: proposed formula immediately before Problem 5.1; §2 for critical equations ([version of record](#); [journal](#)).
- Z. Lai, L.-H. Lim, and K. Ye, *Euclidean Distance Degree in Manifold Optimization*, SIAM Journal on Optimization 35 (2025), 2402–2422, related ED-degree results for flag, Grassmann, and Stiefel models ([journal](#)).

Status check — 2026-09-08

The source computed $D_1 = 4$, $D_2 = 24$, and $D_3 = 544$, and explicitly proposed the displayed pattern. It did not assert a proof. Searches for “symplectic group”, “Euclidean distance degree”, “ED degree”, “544”, and 2024–2026 found no proof, counterexample, or additional general formula. The 2025 related article’s abstract treats different manifold families. No recent explicit reaffirmation of this exact formula was located.

Audit update — 2026-09-10

Rechecked [Baaijens–Draisma](#), §5: the general expression is proposed from small-rank computations, without a proof. Searches for symplectic Euclidean distance degrees and subsequent matrix-manifold ED-degree work found no derivation of this formula. Importance is narrowed to specialist because the requested output is a specific algebraic degree, rather than a general symplectic projection algorithm.